CSC 163: Computing, Ethics, and Society
ABET SO4 · Module A4 (4d)

Societal Effects of Computing Exercises Solution

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Exercise: Classify the effect

For each effect, mark it positive or negative, and the level it acts on: individual, group, society, or environment.

Effect	+ / –	Level
Face unlock saves a user time		
A higher false-match rate for one demographic group		
Constant public surveillance discourages protest		
Data centers consume large amounts of energy and water		
Telehealth gives rural communities access to care		
Deepfakes spread election misinformation		

Exercise Solution: Classify the effect

Name the level by who is affected: one person, a particular group, the public as a whole, or the physical environment.

Effect	+ / –	Level
Face unlock saves a user time	+	**Individual**
A higher false-match rate for one demographic group	–	**Group**
Constant public surveillance discourages protest	–	**Society**
Data centers consume large amounts of energy and water	–	**Environment**
Telehealth gives rural communities access to care	+	**Group**
Deepfakes spread election misinformation	–	**Society**

Exercise: Reading a disparity

A face-matching system runs 20,000 comparisons a day for each of three groups at the same threshold. Compute (a) expected false matches per day for each group and (b) each group's disparity ratio relative to group A. (c) Who needs attention first, and why?

Group	FPR	False matches / day	Ratio to A
A	0.05%		
B	0.5%		
C	1.0%		

Exercise Solution: Reading a disparity

Expected false matches = FPR × comparisons. Example: 0.05% × 20,000 = 10.

(c) Group C: its false matches are 20 times those of group A. Accuracy must be reported **by group**, because an overall average can hide this harm.

Group	FPR	False matches / day	Ratio to A
A	0.05%	**10**	**1×**
B	0.5%	**100**	**10×**
C	1.0%	**200**	**20×**

Exercise: Stakeholders of a hiring tool

A company adopts software that automatically screens resumes. For each stakeholder, name one possible benefit and one possible harm.

Stakeholder	Possible benefit	Possible harm
Applicants		
Groups under-represented in past data		
HR staff		
The company		
Society		

Exercise Solution: Stakeholders of a hiring tool

Include people who did not choose the system (applicants). Evidence: a reported 2018 case where a resume tool trained on past hiring learned to downgrade resumes containing the word 'women's'.

Stakeholder	Possible benefit	Possible harm
Applicants	**Consistent, quick response**	**Qualified people screened out unfairly**
Groups under-represented in past data	**Less individual manager bias, if designed well**	**Past bias reproduced by the model**
HR staff	**Less manual screening**	**Over-trust in the tool; less review**
The company	**Faster hiring, lower cost**	**Legal and reputation risk; missed talent**
Society	**More efficient labor market**	**Discrimination at scale**

Exercise: True or false

Write T or F. If false, correct it with a few words.

Statement	T / F	Correction (if false)
A system's effects fall only on its users		
High overall accuracy means no group is harmed		
Mitigation removes all risk		
A model trained on past hiring data can reproduce past bias		
Biometric data can be reset like a password		
A 4d answer should cover both benefits and harms		

Exercise Solution: True or false

1 point for T/F, 1 point for the correction.

Statement	T / F	Correction (if false)
A system's effects fall only on its users	F	Non-users and society are affected too
High overall accuracy means no group is harmed	F	Error rates can differ sharply by group; measure per group
Mitigation removes all risk	F	Residual risk remains and should be stated
A model trained on past hiring data can reproduce past bias	T	—
Biometric data can be reset like a password	F	It cannot be changed after a breach
A 4d answer should cover both benefits and harms	T	—

Exercise: Practice 4d question

Discuss the **societal effects** of a ride-sharing platform: who benefits, who is harmed, and what you would do about it. (Target: about half a page.)

Exercise Solution: Practice 4d question

- **Model answer.**
- Solution: an app matches riders with independent drivers.
- Positive: affordable, convenient rides; flexible income for drivers; mobility where taxis are scarce.
- Negative: disputes over driver pay and benefits (worker classification); added congestion; pressure on taxi drivers; riders' location data collected.
- Weigh and mitigate: clear pay and safety standards, limits on data use and retention, accessibility options; cite ACM 1.1 (benefit to society), 1.2 (avoid harm), 1.6 (privacy). Residual risk: platform power remains.
- **Scoring (4 points):** several effects on several groups (1); benefits and harms both covered (1); evidence or a concrete example (1); balance, mitigation, and a professional-ethics reference (1).